

Dataset compression with a trained sparse basis

A real byte-level codec — transform, top- k , quantise, entropy-code, store, decode — demonstrating full-dataset recovery from $\leq 50\%$ of the raw bytes

Setup

Each image is stored as an actual file: coefficients under a basis, top- k by magnitude, symmetric uniform b -bit quantisation, a kept-position bitmask, zlib. The decoder reconstructs from the file alone plus the basis parameter file (stored once per dataset and counted in every size below). The codec is identical across bases; only the transform differs. Grid: $k/d \in \{0.05, \dots, 0.5\}$, $b \in \{6, 8, 10\}$, all 550 images (500 train + 50 held-out test, seed-42 split). All PSNR/SSIM below are on the 50 held-out test images. The real-valued rich bases were retrained at the headline budget (1008 steps, no early stop) because the original runs saved only metrics, not parameters (see “Reproduction gate” below).

Headline: recovery at half the bytes

At a total budget of 50% of raw uint8 size, the best operating points are:

dataset / basis	B/img	% raw	test PSNR (dB)	test SSIM
QuickDraw — real rich (trained)	494.8	48.3%	44.69	0.99
QuickDraw — block DCT 8×8	484.6	47.3%	36.88	0.97
DIV2K-8q — real rich (trained)	31868.3	48.6%	40.67	0.98
DIV2K-8q — block DCT 8×8	31428	48%	40.96	0.98

On QuickDraw the trained real-valued rich basis recovers the dataset at 48.3% of its raw size with a 7.81 dB PSNR advantage over the identical codec built on block DCT. On DIV2K the two are at parity (-0.29 dB), consistent with the committed top- k metrics.

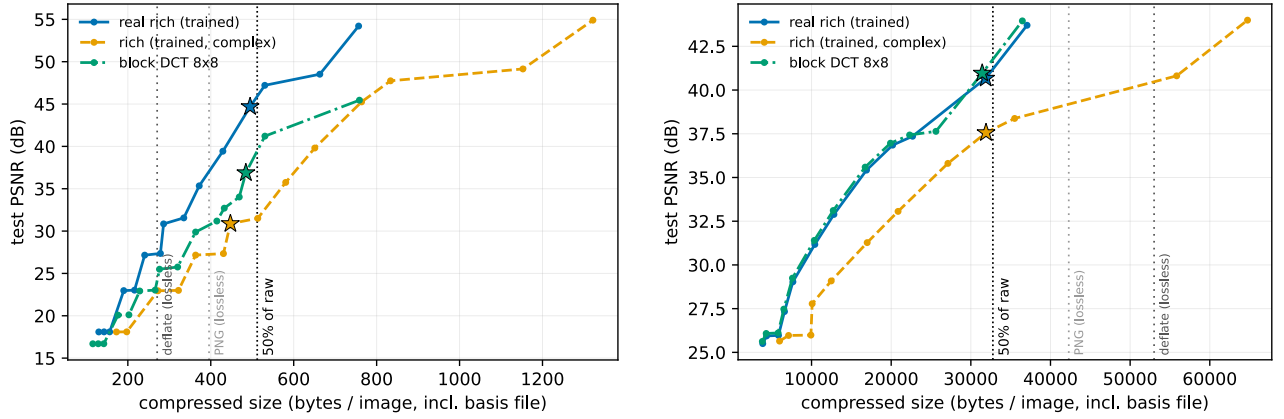


Figure 1: Rate-distortion in real bytes per image (blob + amortised basis file). Left: QuickDraw. Right: DIV2K-8q. Stars mark the best operating point within the 50%-of-raw budget (dotted vertical). Grey dotted verticals: lossless deflate / PNG references.

What the advantage measures

Per image, with pixels in $[0, 1]$ (peak level $L = 1$) and the decoded image clamped to that range,

$$\text{MSE}(x, \hat{x}) = \frac{1}{HW} \sum_{i,j} (x_{ij} - \hat{x}_{ij})^2, \quad \text{PSNR} = 10 \log_{10} \left(\frac{L^2}{\text{MSE}} \right) = -10 \log_{10} \text{MSE} \text{ [dB]}.$$

Reported values are the means of the per-image metrics over the 50 held-out test images.

PSNR advantage. The logarithm turns error ratios into differences, so the advantage at the common byte budget B (here 50% of raw) is a multiplicative error statement:

$$\Delta_{\text{PSNR}} = \text{PSNR}_{\text{rich}}(B) - \text{PSNR}_{\text{DCT}}(B) = 10 \log_{10} \left(\frac{\text{MSE}_{\text{DCT}}}{\text{MSE}_{\text{rich}}} \right).$$

QuickDraw’s $\Delta_{\text{PSNR}} = 7.81$ dB therefore says that at equal storage the block-DCT reconstruction carries $6 \times$ the squared error ($2.46 \times$ the RMS pixel error, $10^{\Delta/20}$) of the trained basis. One aggregation subtlety: since the table reports the mean of per-image decibel values, the ratio is between **geometric** means of per-image MSE across the test split, not arithmetic means. DIV2K’s $\Delta_{\text{PSNR}} = -0.29$ dB is a factor 0.94 — parity.

SSIM (Wang et al. 2004; scikit-image with data range 1: 7×7 windows, $C_1 = (0.01L)^2$, $C_2 = (0.03L)^2$) scores window-local agreement in luminance, contrast and structure, reaching 1 only for identical images:

$$\text{SSIM}(x, \hat{x}) = \frac{1}{M} \sum_{w=1}^M \frac{(2\mu_x \mu_{\hat{x}} + C_1)(2\sigma_{x\hat{x}} + C_2)}{(\mu_x^2 + \mu_{\hat{x}}^2 + C_1)(\sigma_x^2 + \sigma_{\hat{x}}^2 + C_2)},$$

with means μ , variances σ^2 and covariance $\sigma_{x\hat{x}}$ taken over each sliding window w . Because scores sit close to 1, the informative quantity is the **residual structural distortion** $D = 1 - \text{SSIM}$, and the advantage is its ratio at matched bytes:

$$\frac{D_{\text{DCT}}}{D_{\text{rich}}} = \frac{1 - \text{SSIM}_{\text{DCT}}}{1 - \text{SSIM}_{\text{rich}}}.$$

On QuickDraw $D_{\text{DCT}}/D_{\text{rich}} \approx 4.1$ — block DCT retains about $4.1 \times$ the structural distortion of the trained basis at the same stored size. On DIV2K the ratio is 0.92 — parity, matching the PSNR picture.

Honest accounting

Complex coefficients cost double. The complex rich basis stores re + im per kept coefficient, so at matched bytes it operates at half the coefficient count — visible as the orange curve sitting below the real-valued blue one on both datasets (QuickDraw: 30.88 dB; DIV2K: 37.56 dB at the 50% budget). This is why the real-valued variant is the storage contender.

Lossless floor. QuickDraw sketches are mostly background: lossless deflate of the raw bytes is already down to $\sim 26\%$ of raw size, and per-image PNG sits at $\sim 39\%$ — both vertical references fall well left of the 50% line in the figure, comfortably below any of the trained-basis operating points shown. The QuickDraw claim is therefore basis-versus-basis at a matched lossy budget, not versus-PNG. DIV2K natural images have no such floor (deflate $\sim 81\%$, PNG $\sim 65\%$ of raw) and carry the size claim directly.

DIV2K parity. On DIV2K the trained basis matches block DCT rather than beating it, consistent with the committed top- k metrics; the win case is QuickDraw.

Reproduction gate

DIV2K retraining reproduced the committed structure-tree cells exactly (≤ 0.01 dB at every keep ratio, both bases). QuickDraw did not: absolute PSNR sits 0.7–1.0 dB below the committed cells because the QuickDraw .npy snapshot on this machine differs from the one the committed run used — even the training-free classical baselines shift (deterministic transforms, same nominal seed-42 split), while the real_rich — block_dct_8 **gap** reproduces within 0.2 dB at every keep ratio. Details in quickdraw_5q/checkpoints/GATE_NOTE.md. All numbers in this section are therefore within-run comparisons on the current data snapshot.